**DATABASE MANAGEMENT SYSTEMS (IT 2040)**

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**Question 1**

What are the three main objectives of database security?

- A) Confidentiality, Integrity, Authentication
- B) Secrecy, Integrity, Availability
- C) Privacy, Security, Accessibility
- D) Authorization, Authentication, Auditing

**Answer: B**

---

**Question 2**

Which security objective ensures that information should not be disclosed to unauthorized users?

- A) Integrity
- B) Availability
- C) Secrecy
- D) Authentication

**Answer: C**

---

**Question 3**

Which security objective ensures that only authorized users can modify data?

- A) Secrecy
- B) Integrity
- C) Availability
- D) Confidentiality

**Answer: B**

---

**Question 4**

Which security objective ensures that authorized users are not denied access?

- A) Secrecy
- B) Integrity
- C) Availability

- D) Reliability

---

**Answer: C**

**Question 5**

What is a security policy?

- A) A backup strategy
- B) A document specifying who is authorized to do what
- C) A database design document
- D) A disaster recovery plan

**Answer: B**

---

**Question 6**

What are the two main access control mechanisms in DBMS?

- A) User control and Admin control
- B) Discretionary access control and Mandatory access control
- C) Role-based and Permission-based control
- D) Public and Private access control

**Answer: B**

---

**Question 7**

What is Discretionary Access Control (DAC) based on?

- A) User roles only
- B) Security clearance levels
- C) Access rights or privileges
- D) Time-based access

**Answer: C**

---

**Question 8**

What is authentication in SQL Server security?

- A) Granting permissions to users
- B) Verifying the identities of users trying to connect

C) Creating user accounts

**Answer: B**

D) Monitoring user activities

---

**Question 9**

What is authorization in SQL Server?

- A) Creating user logins
- B) Verifying user identity
- C) Determining which resources authorized users can access and what actions they can take
- D) Encrypting data

**Answer: C**

---

**Question 10**

What are Principals in SQL Server security?

- A) Database objects like tables
- B) Individuals, groups, or processes granted access to SQL Server
- C) Types of permissions
- D) Server configurations

**Answer: B**

---

**Question 11**

What are server-level principals?

- A) Database users and roles
- B) Tables and views
- C) Logins and server roles
- D) Stored procedures

**Answer: C**

---

**Question 12**

What are database-level principals?

- A) Logins and server roles
- B) Users and database roles
- C) Tables and views
- D) Schemas and catalogs

**Answer: B**

---

**Question 13**

What are Securables in SQL Server?

- A) Users and logins
- B) Objects that make up the server and database environment
- C) Permissions granted to users
- D) Authentication methods

**Answer: B**

---

**Question 14**

What are the three hierarchical levels of securables?

- A) User, Group, Admin
- B) Read, Write, Execute
- C) Server, Database, Schema
- D) Public, Private, Protected

**Answer: C**

---

**Question 15**

What are Permissions in SQL Server?

- A) User accounts
- B) Types of access permitted to principals on specific securables
- C) Database objects
- D) Server configurations

**Answer: B**

---

**Question 16**

What is a Login in SQL Server?

- A) A database-level user account

**Answer: B**

---

- B) A server-level individual user account for logging into SQL Server instance

C) A permission type

D) A database object

**Question 17**

How many authentication modes does SQL Server support?

- A) 1
- B) 2
- C) 3
- D) 4

**Answer: B**

---

**Question 18**

Which authentication mode uses Windows user or group accounts?

- A) SQL Server Authentication
- B) Windows Authentication
- C) Mixed Authentication
- D) Certificate Authentication

**Answer: B**

---

**Question 19**

What is the syntax to create a Windows Authentication login?

- A) CREATE LOGIN [username] FROM WINDOWS
- B) CREATE LOGIN [username] WITH PASSWORD
- C) CREATE WINDOWS LOGIN [username]
- D) ADD LOGIN [username] WINDOWS

**Answer: A**

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**Answer: B**

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**Question 20**

What is required when creating a SQL Server Authentication login?

- A) Windows domain name
- B) Password

C) Email address

D) Security certificate

**Question 21**

What does MUST_CHANGE option do when creating a login?

- A) User must change password every month
- B) User must change password on first login
- C) Password changes automatically
- D) User cannot change password

**Answer: B**

---

**Question 22**

What is the difference between a Login and a User?

- A) No difference, they are synonyms
- B) Login is server-level for connection, User is database-level for permissions
- C) Login is for admins, User is for regular users
- D) Login is temporary, User is permanent

**Answer: B**

---

**Question 23**

What is a server role?

- A) A database user

**Answer: B**

---

- B) A group of logins that share common server-level permissions
- C) A type of securable
- D) A permission type

**Answer: B**

---

**Question 24**

What is the difference between fixed and user-defined server roles?

A) No difference

B) Fixed roles have predefined permissions (can't change), user-defined roles have customizable permissions

- C) Fixed roles are temporary
- D) User-defined roles require more memory

**Answer: B**

---

**Question 25**

Which fixed server role can perform any activity in the server?

- A) dbcreator
- B) securityadmin
- C) sysadmin
- D) serveradmin **Answer: C**

---

**Question 26**

Which fixed server role can create, alter, drop, and restore databases?

- A) sysadmin
- B) dbcreator
- C) databaseadmin
- D) serveradmin

**Answer: B**

---

**Question 27**

Which fixed server role manages logins and their properties?

- A) loginadmin
- B) securityadmin
- C) useradmin
- D) sysadmin

**Answer: B**

---

**Question 28**

How do you add a login to a fixed server role?

- A) ADD LOGIN [login] TO ROLE [role]
- B) ALTER SERVER ROLE [role] ADD MEMBER [login]
- C) GRANT ROLE [role] TO [login]

- D) INSERT INTO [role] VALUES ([login])

---

**Answer: B**

---

### Question 29

How do you create a user-defined server role?

- A) ADD SERVER ROLE [name]
- B) CREATE ROLE [name]
- C) CREATE SERVER ROLE [name]
- D) NEW SERVER ROLE [name]

**Answer: C**

---

### Question 30

What permission allows creating databases on the server?

- A) DATABASE CREATE
- B) CREATE ANY DATABASE
- C) DB_CREATE
- D) MAKE DATABASE

**Answer: B**

---

### Question 31

What is a database user?

- A) Same as a login
- B) A database-level account mapped to a login
- C) A server role
- D) A permission type

**Answer: B**

---

### Question 32

What is the syntax to create a database user?

- A) CREATE USER [user] WITH LOGIN [login]

□

□

□

□

- B) ADD USER [user] FROM LOGIN [login]
- C) CREATE USER [user] FOR LOGIN [login]
- D) NEW USER [user] AS [login]

**Answer: C**

**Question 33**

Can one login have users in multiple databases?

- A) No, one login = one user only
- B) Yes, one login can be mapped to users in multiple databases
- C) Only for sysadmin logins
- D) Only with special permissions **Answer: B**

---

**Question 34**

What is a database role?

- A) A server-level group
- B) A group of database users sharing common database-level permissions
- C) A type of table
- D) A backup strategy

**Answer: B**

---

**Question 35**

Which fixed database role can perform all configuration and maintenance activities on the database?

- A) db_admin
- B) db_owner
- C) db_maintainer
- D) db_sysadmin

**Answer: B**

---

**Question 36**

Which fixed database role can run any DDL command in a database?

A) db_owner

B) db_admin

C) db_ddladmin

D) db_creator

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Answer: C

### Question 37

Which fixed database role can add or remove database access?

- A) db_accessadmin
- B) db_useradmin
- C) db_securityadmin
- D) db_gatekeeper

**Answer: A**

### Question 38

How do you create a user-defined database role?

- A) ADD ROLE [name]
- B) CREATE ROLE [name]
- C) NEW DATABASE ROLE [name]
- D) CREATE DATABASE ROLE [name]

**Answer: B**

### Question 39

What does the GRANT command do?

- A) Creates a new permission
- B) Grants permissions on a securable to a principal
- C) Removes permissions
- D) Lists all permissions

**Answer: B**

### Question 40

What does the DENY command do?

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- 
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- 

A) Removes permissions

B) Explicitly denies a permission to a principal

C) Grants negative permissions

D) Disables a user account

**Answer: B**

**SECTION B: MULTIPLE ANSWER QUESTIONS (Select ALL that apply) - 15 Questions**

**Question 41**

The three main objectives of database security are: **(Select all that apply)**

- A) Secrecy - prevent unauthorized disclosure
- B) Speed - fast query execution
- C) Integrity - prevent unauthorized modification
- D) Availability - ensure authorized access
- E) Simplicity - easy to use
- F) Scalability - handle growth

**Answers: A, C, D**

---

**Question 42**

Which are components of SQL Server security framework? **(Select all that apply)**

- A) Principals
- B) Securables
- C) Permissions
- D) Databases
- E) Tables
- F) Authentication

**Answers: A, B, C, F**

---

**Question 43**

Server-level principals include: **(Select all that apply)**

- A) Logins
- B) Server roles

C) Database users

D) Database roles

E) Windows groups mapped to logins

F) Applications

**Answers: A, B**

**Question 44**

Database-level principals include: **(Select all that apply)**

- A) Logins
- B) Server roles
- C) Database users
- D) Database roles
- E) Application roles
- F) Schemas

**Answers: C, D, E**

---

**Question 45**

Server-level securables include: **(Select all that apply)**

- A) Databases
- B) Tables
- C) Availability groups
- D) Views
- E) Endpoints
- F) Logins

**Answers: A, C, E, F**

---

**Question 46**

Schema-level securables include: **(Select all that apply)**

- A) Databases
- B) Tables
- C) Views
- D) Functions
- E) Stored procedures
- F) Schemas

**Answers: B, C, D, E**

□

□

□

□

**Question 47**

Authentication modes in SQL Server include: **(Select all that apply)**

- A) Windows Authentication
- B) SQL Server Authentication
- C) Certificate Authentication
- D) Mixed Mode (both Windows and SQL)
- E) Kerberos Authentication
- F) LDAP Authentication

**Answers: A, B, D**

---

**Question 48**

When creating a SQL Server Authentication login, you can specify: **(Select all that apply)**

- A) Password
- B) MUST_CHANGE - force password change on first login
- C) CHECK_EXPIRATION - enable password expiration
- D) CHECK_POLICY - enforce password policy
- E) DEFAULT_DATABASE
- F) DEFAULT_SCHEMA

**Answers: A, B, C, D, E**

---

**Question 49**

Fixed server roles include: **(Select all that apply)**

- A) sysadmin - perform any activity
- B) dbcreator - create/manage databases
- C) securityadmin - manage logins
- D) serveradmin - configure server settings
- E) diskadmin - manage disk files
- F) db_owner - database owner

**Answers: A, B, C, D, E**

---

**Question 50**

Fixed database roles include: **(Select all that apply)**

- A) db_owner - all database permissions
- B) db_securityadmin - manage permissions
- C) db_accessadmin - manage access
- D) db_ddladmin - DDL operations
- E) db_datareader - read all tables
- F) db_datawriter - modify all tables

**Answers: A, B, C, D, E, F**

---

**Question 51**

Permissions that can be granted on database objects include: **(Select all that apply)**

- A) SELECT - read data
- B) INSERT - add data
- C) UPDATE - modify data
- D) DELETE - remove data
- E) EXECUTE - run procedures/functions
- F) REFERENCES - create foreign keys

**Answers: A, B, C, D, E, F**

---

**Question 52**

Server-level permissions include: **(Select all that apply)**

- A) CREATE ANY DATABASE
- B) ALTER ANY LOGIN
- C) SELECT - read tables
- D) SHUTDOWN - stop server
- E) VIEW SERVER STATE
- F) CREATE TABLE

**Answers: A, B, D, E**

---

**Question 53**

Advantages of using roles include: **(Select all that apply)**

- A) Easier permission management
- B) Centralized permission control
- C) Less administrative overhead
- D) Improved performance
- E) Group users with similar needs
- F) Automatic user creation

**Answers: A, B, C, E**

---

**Question 54**

GRANT command can: **(Select all that apply)**

- A) Grant permissions to users
- B) Grant permissions to roles
- C) Include WITH GRANT OPTION
- D) Remove permissions
- E) Deny permissions
- F) Grant multiple permissions in one statement

**Answers: A, B, C, F**

---

**Question 55**

Views can be used for security by: **(Select all that apply)**

- A) Hiding sensitive columns
- B) Filtering rows based on criteria
- C) Presenting summary information
- D) Encrypting data automatically
- E) Providing read-only access to data
- F) Combining GRANT/REVOKE for access control

**Answers: A, B, C, E, F**

---

**SECTION C: SCENARIO-BASED & ANALYTICAL QUESTIONS - 5 Questions**

**Question 56**

**Scenario:** Super market project with Senior DBA (Kamal), Junior DBAs (Namali, Janaka), Developer (Sahan), and Data Entry Operators (Amali, Mihiran).

**Senior DBA needs to create databases and logins. Which approach is MOST secure?**

A) Project Manager creates all logins, assigns sysadmin to everyone

B) Project Manager creates Senior DBA login with dbcreator and securityadmin roles, Senior DBA creates other logins

C) All users get sysadmin role

D) Everyone uses the same login

**Answer: B**

---

**Question 57**

**Code Analysis - Creating Login:**

```
CREATE LOGIN namali

WITH PASSWORD = 'namali@123' MUST_CHANGE,

CHECK_EXPIRATION = ON,

CHECK_POLICY = ON,

DEFAULT_DATABASE = Inventory;
```

**What security features does this implement?**

- A) Forces password change on first login only
- B) Forces password change + enables expiration + enforces complexity policy
- C) Creates Windows Authentication login
- D) Grants sysadmin privileges

**Answer: B**

---

**Question 58**

**Scenario:** Junior DBA (Namali) needs to manage InventoryDB - create tables, modify schema, manage users, and run backups.

**Which roles should be assigned?**

```
-- Option A

ALTER ROLE sysadmin ADD MEMBER namali


-- Option B

USE InventoryDB

ALTER ROLE db_owner ADD MEMBER namali
```

-- Option C

USE InventoryDB

ALTER ROLE db_owner ADD MEMBER namali

ALTER ROLE db_securityadmin ADD MEMBER namali

ALTER ROLE db_accessadmin ADD MEMBER namali

-- Option D

USE InventoryDB

ALTER ROLE db_ddladmin ADD MEMBER namali

ALTER ROLE db_backupoperator ADD MEMBER namali **Answer:**

**B**

---

**Question 59**

**Scenario:** Database Developer (Sahan) needs to create tables with foreign keys but shouldn't access production data or manage users.

**Which permissions are appropriate?**

-- Option A

CREATE ROLE databaseDev

GRANT CREATE TABLE, ALTER, REFERENCES ON DATABASE::InventoryDB TO databaseDev

ALTER ROLE databaseDev ADD MEMBER Sahan

-- Option B

ALTER ROLE db_owner ADD MEMBER Sahan

-- Option C

CREATE ROLE databaseDev

GRANT SELECT, INSERT, UPDATE, DELETE, CREATE TABLE ON DATABASE::InventoryDB TO databaseDev

ALTER ROLE databaseDev ADD MEMBER Sahan

-- Option D

ALTER ROLE db_ddladmin ADD MEMBER Sahan

**Answer: A**

**Question 60**

**Security Policy Design:**

**Requirements:**

- Data Entry Operators: Insert and view data only in specific tables (Products, Inventory)
- Analysts: Read all tables, cannot modify
- Managers: Full access to all data, grant permissions to others
- Audit requirement: Track who modified what **Which implementation is BEST?**

```
-- Option A
CREATE ROLE DataEntry
GRANT SELECT, INSERT ON Products TO DataEntry
GRANT SELECT, INSERT ON Inventory TO DataEntry
-- Add members

CREATE ROLE Analyst
GRANT SELECT ON DATABASE::InventoryDB TO Analyst
-- Add members

ALTER ROLE db_owner ADD MEMBER ManagerUser
-- Implement triggers for auditing

-- Option B
Everyone gets sysadmin role
-- Use application-level controls

-- Option C
CREATE ROLE DataEntry
ALTER ROLE db_datawriter ADD MEMBER DataEntry  -- All tables

CREATE ROLE Analyst
ALTER ROLE db_datareader ADD MEMBER Analyst
```

CREATE ROLE Manager

ALTER ROLE db_owner ADD MEMBER Manager

GRANT ALTER ANY USER TO Manager WITH GRANT OPTION

**Answer: A**

---

**Question 61**

Supermarket item table stores itemNumber, description, itemPrice, quantity available, and reorder level. Which of the following are true? (Select one or more)
• A) DBMS cannot allow multiple cashiers to update the quantity in hand simultaneously.
• B) DBMS can be configured so registered customers can only view item descriptions and prices.
• C) DBMS can be configured so that only the manager can change the price of an item.
• D) DBMS cannot allow multiple cashiers to access the table to see the price of items at the same time.
• E) DBMS can be configured so that item numbers cannot be duplicated.

**Answer: B, C, E**

---

**Question 62**

Consider activities performed by a database developer for a small pharmacy:
1. Go through books recording supplies.
2. Identify attributes that determine certain groups of attributes.
3. Select a database development software.
4. Give access to clerks to enter data.
Select the correct sequence:
• A) 1, 2, 3, 4
• B) 2, 3, 4, 1
• C) 4, 1, 3, 2
• D) 1, 3, 2, 4
• E) 3, 2, 4, 1

**Answer: D**

---

**Question 63**

Which of the following is true with respect to the item table?
• A) DBMS can be configured in a manner that item numbers cannot be duplicated
• B) DBMS can be configured in a manner that registered customers can only view the item descriptions and prices

• C) DBMS cannot allow multiple cashiers to access the table to update the quantity in hand
• D) DBMS cannot allow multiple cashiers to access the table to see the price of items at the same time
• E) DBMS can be configured in a manner that only the manager can change the price of an item

**Answer: A, B, E**

---

**Question 64**

Steps involved in database design process for a bank database:
1. Collecting printed reports presented at meetings.
2. Select a database software to develop the database.
3. Providing access to senior managers to change interest rate of an account type.
Correct order for designing and developing the database:
• A) 4, 2, 3, 1
• B) 3, 4, 2, 1
• C) 1, 2, 3, 4
• D) 2, 3, 1, 4
• E) 3, 2, 4, 1

**Answer: C**

---

**Question 65**

An item table of a supermarket stores the itemNumber, description, item price, quantity available, and reorder level. Which of the following is true with respect to the above table?
• A) DBMS can be configured in a manner that item numbers cannot be duplicated
• B) DBMS can be configured in a manner that registered customers can only view the item descriptions and prices
• C) DBMS cannot allow multiple cashiers to access the table to update the quantity in hand
• D) DBMS cannot allow multiple cashiers to access the table to see the price of items at the same time
• E) DBMS can be configured in a manner that only the manager can change the price of an item

**Answer: A, B, E**

---

**Question 66**

Database developer – Requirement collection Intensions of a database developer:
• A) Identify different types of data retrievals
• B) Identify the number of concurrent users
• C) Find relationships among data in the organization
• D) Find names of the people who will develop applications
• E) Find data to be stored in the organization

**Answer: A, B, C, E**

---

Question 67

Database developer tasks (pharmacy)
1. Go through books for supplies
2. Identify attributes that determine certain groups of attributes
3. Select a database development software
4. Give access to clerks to enter data
Correct order:
• A) 3, 2, 4, 1
• B) 1, 2, 3, 4
• C) 2, 3, 4, 1
• D) 1, 3, 2, 4
• E) 4, 1, 3, 2

**Answer: D**

---

**Question 68**

Consider the following steps involved in the database design process for a bank database:
1. Develop a database program to calculate the interest
2. Collect printed reports presented at meetings
3. Select a database software to develop the database
4. Provide access to senior managers to change the interest rate of an account type
Which order should the above steps happen in designing and developing a database?
• A) 2, 3, 1, 4
• B) 1, 2, 3, 4
• C) 3, 2, 4, 1
• D) 4, 2, 3, 1
• E) 3, 4, 2, 1

**Answer: A**

---

**Question 69**

Which of the following situations would you use a database to store data? (Select one or more)
• A) To store information about rooms and customers of a hotel management system
• B) To store a student name list of your class which is used by multiple lecturers
• C) To store information of vehicles owned by a vehicle renting company
• D) To store your 'to do' list
• E) To store addresses of relatives and friends

**Answer: A, B, C**

---

**Question 70**

Which of the following is/are the intention(s) of a database developer during the requirement collection and analysis phase? (Select one or more)
• A) Identify different types of data retrievals to be performed on the database
• B) Find the names of the people who will be developing the applications to access the database
• C) Identify the number of concurrent users who will be using the database
• D) Find data to be stored in the organization
• E) Find relationships among data in the organization

**Answer: A, C, D, E**

---

**Question 71**

An item table of a supermarket stores the itemNumber, description, item Price, Quantity available, and re-order level. Which of the following is true with respect to the above table? (Select one or more)
• A) DBMS cannot allow multiple cashiers to access the table to update the quantity in hand
• B) DBMS can be configured in a manner that only the manager can change the price of an item
• C) DBMS can be configured in a manner that item numbers cannot be duplicated
• D) DBMS can be configured in a manner that registered customers can only view the item descriptions and prices
• E) DBMS cannot allow multiple cashiers to access the table to see the price of items at the same time

**Answer: B, C, D**

---

**Question 72**

Which of the following is/are intentions of a database developer during the requirement collection and analysis? (Select one or more)
• A) Identify different types of data retrievals to be performed on the database
• B) Finding the names of the people who will be developing the applications to access the database
• C) Identify the number of concurrent users who will be using the database
• D) Finding data to be stored in the organization
• E) Finding relationships among data in the organization

**Answer: A, C, D, E**

---